

Parsa Rangriz

CONTACT INFORMATION	Department of Mathematics, UC San Diego, 9500 Gilman Drive, La Jolla, CA, USA, 92093 Office: HSS 4047	prangriz@ucsd.edu https://prangriz.github.io
EDUCATION	University of California, San Diego PhD in Mathematics University of Waterloo MMath in Statistics <i>Advisor:</i> Aukosh Jagannath <i>Thesis:</i> High-dimensional scaling limits of online SGD in single-index models Sharif University of Technology BSc in Physics <i>Extra:</i> Minor in Mathematics	2025 - Present 2023 - 2025 2018 - 2023
PUBLICATIONS	Parsa Rangriz, <i>Limit theorems for stochastic gradient descent in high-dimensional single-layer networks</i> , Electronic Communications in Probability, 31, No. 15 (2026). Parsa Rangriz, <i>High-dimensional scaling limits of online stochastic gradient descent in single-index models</i> , MMath Thesis, University of Waterloo (2025).	
EMPLOYMENT	Graduate Research Assistant. University of Waterloo <i>Supervisor:</i> Aukosh Jagannath Worked on high-dimensional probability, scaling limits, random matrices, and mathematical foundations of machine learning. Summer Research Intern. EPFL <i>Supervisor:</i> Lenka Zdeborova Worked on statistical physics methods for machine learning and combinatorial optimization, e.g., mean-field theory, replica method, and message-passing algorithms.	2023 - 2025 2022
AWARDS & HONORS	First-Year Departmental Fellowship. UC San Diego Graduate Research Studentship. University of Waterloo International Master’s Award of Excellence. University of Waterloo Master of Math Entrance Scholarship. University of Waterloo Summer@EPFL Fellowship. EPFL Silver Medal in the Iranian National Physics Olympiad	2025 - 2026 2023 - 2025 2023 2023 2022 2017
TA EXPERIENCE	UC San Diego MATH 18: Linear Algebra (Fall 2026); PHYS 1CL: Waves, Optics & Modern Physics Lab (Summer 2026); MATH 193A: Actuarial Mathematics (Summer 2026); MATH 180A: Introduction to Probability (Spring 2026); MATH 10B: Calculus II (Winter 2026); MATH 11: Calculus-Based Probability and Statistics (Fall 2025) University of Waterloo STAT 333: Stochastic Processes I (Winter 2025, Winter 2024); STAT 433: Stochastic Processes II (Fall 2024); STAT 330: Mathematical Statistics (Fall 2023, Spring 2024); STAT 231: Probability (Spring 2024); STAT 230: Statistics (Spring 2024); STAT 202: Introductory Statistics for Scientists (Fall 2023)	2025 - Present 2023 - 2025